

Pavle Petrović

Principal Software Engineering Manager @ Tenstorrent

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Based in Belgrade, Serbia

Profile

Principal software engineer and technical leader with 15+ years of experience in machine learning and game development. At Ubisoft, shipped three AAA games and long-running live updates, followed by work on the first player-facing machine learning framework for bots used across multiple titles. Currently working on LLM inference deployments on Tenstorrent hardware.

Leadership Summary

Engineering leader with people management responsibility for senior engineers and technical leads. Own technical direction, architecture, and delivery of customer-facing products and shared libraries used across engineering teams. Hiring, onboarding, team scaling, and customer-facing technical ownership.

Machine Learning & Inference Optimizations

Tenstorrent, Principal Software Engineering Manager

Product Software Engineering, Jan 2025 - ONGOING

- Lead the LLM inference team in Product Software Engineering, growing it from a single-contributor role to 6 engineers during a period of rapid organizational growth.
- Drive LLM inference optimization on Tenstorrent's open-source software stack, owning latency, throughput, accuracy, quantization, prompting features, and stability for customer deployments.
- Lead the Gemma-3 initiative from initial bring-up through open-source collaboration to a fully functional model using vLLM v1, enabling a successful PoC for 2 enterprise customers.
- Accountable for the data-parallel implementation and 2x throughput improvement of Llama-3.1-8B under concurrent user load.
- Take flagship models, including Llama-3.3-70B, from model-complete to product-ready, driving CI validation, feature completeness, and customer-facing releases.
- Work directly with customers on requirements, integration, and customer-specific features.

Ubisoft, Senior Technical Lead

Machine Learning Framework, Feb 2022 - Dec 2024

- Led development of Ubisoft's first player-facing machine learning framework integrated into in-house engines, shipping gameplay features (navigation, combat) in Rainbow Six Siege, For Honor, and Star Wars Outlaws, and integrating ML into animation pipelines and gameplay testing across Assassin's Creed, Far Cry, and other titles.
- Owned architecture, technical roadmap, and delivery of core ML

framework components, driving adoption across teams, games, and engines while balancing performance, usability, and long-term maintainability.
- Led a geo-distributed senior engineering team, combining hands-on technical leadership with people management.
- Played a key role in defining a company-wide machine learning pipeline spanning data collection, training and in-game integration, focusing on technical feasibility and adoption constraints.
- Worked directly with game engineering teams, research (Ubisoft La Forge), and production leadership to align technical direction and drive framework adoption.

AAA Game Development & Real-Time Systems

Ubisoft, Technical Lead

Riders Republic, Jul 2019 - Feb 2022

- Technical lead for PC and cloud platforms (Stadia, Amazon Luna) on a large-scale open-world multiplayer title.
- Defined platform architecture and standards, including hardware requirements, rendering APIs, input/UI specialization, security, and performance.
- Owned technical direction and hands-on implementation for systemic gameplay systems and PvP modes used by a multi-million player base.
- Provided technical leadership for 20+ engineers across gameplay, engine, and rendering teams.
- Primary technical point of contact with the lead studio and external partners (Nvidia, AMD, Google).
- Improved cross-studio development processes, resulting in increased trust and greater technical ownership for the local studio and team.

Ubisoft, Lead Programmer

Steep & Ghost Recon Wildlands, Oct 2015 - Jul 2019

- Led programming teams across gameplay, UI, engine, and rendering on AAA open-world titles.
- Owned PC platform quality, performance, and feature development across live and production phases.
- Played a key role in scaling the Ubisoft Belgrade studio from 9 to ~200 engineers, contributing to hiring, onboarding, and team structure.
- Established development standards and knowledge-sharing practices to support rapid studio growth.
- Represented the studio in cross-site technical coordination and collaboration.

Early Career: Embedded Systems & Low-Level Software

Intel, Imaging Firmware Engineer

Mobile Communications Group, Mar 2013 - Jun 2015

HTEC Group, DSP / Embedded Systems Engineer

Embedded Electronics, May 2010 - Mar 2013

- Developed imaging algorithms and performance-critical firmware on custom ISP/DSP hardware, working close to the metal on VLIW architectures using C++, OpenCL, and MATLAB.
- Optimized image quality and latency under tight hardware and real-time constraints, including integration into the Android camera stack across multiple system layers.

Education

Electrical and Computer Engineering / MSc & BSc

University of Belgrade, 2004 - 2010

Digital Arts and New Media / MA

Faculty for Media and Communications, 2014 - 2015

Skills & Tools

Technical

C++, Python, LLM inference pipelines, LLM runtimes, shared multi-team codebases, large game engine codebases, software architecture, API design, performance optimization, memory management, profiling, debugging

Leadership & Delivery

Technical direction, people management, hiring, onboarding, risk assessment, trade-off decisions, cross-team collaboration, customer-facing technical ownership

Tools

Git, Perforce, VS Code, Visual Studio, Jira, Confluence, Miro

Talks & Conferences

Data Science Conference, Belgrade (2023)
Talk: Experimenting or Delivering, Stories About Industrializing In-Game Machine Learning Solutions

Ubisoft Developers Conference, Montreal (2023)
Talk: Machine Learning Framework

Languages

English (Fluent), Serbian (Native)

Art & Creative Practice

Digital and interactive art, including internationally exhibited large-scale light installations combining software, hardware, and light.

Making music and performing live, released an EP and a full-length album on streaming platforms.

Live visual performance (VJing) for music and cultural events.